Genotype and Phenotype Lab

1. Purpose and Objectives

The purpose of this investigation is to determine your own genotype for the taste receptor gene TAS2R38 and correlate the genotype with the phenotype, that is the ability to taste phenylthiocarbamide (PTC).

After completing this lab, you should be able to:

- Understand the difference between phenotype and genotype.
- Describe how DNA (deoxyribonucleic acid) sequence (genotype) determines an organism's characteristics (phenotype)
- Observe how a change in DNA sequence may alter phenotype
- Explain how polymerase chain reaction (PCR) can be used to produce copies of a selected segment of DNA
- Explain how restriction fragment length polymorphism can identify differences in genotype
- Describe how DNA gel electrophoresis works.
- Use a bioinformatics resource tool to obtain and explore gene sequences.

2. Safety Guidelines

- All samples must be disposed of in autoclave bags for autoclaving prior to disposal.
- Handle the pipettmen with care. Do not get any liquid on or into any pipettman.
- Always wear safety goggles when using chemicals in the lab.
- Clean and replace all lab equipment. Clean your table when finished.
- Wash your hands before leaving the lab.

Background

Recall that **DNA** (**deoxyribonucleic acid**) is the genetic material found in cells. The nucleotide sequence in DNA encodes information responsible for determining the characteristics of living things. DNA belongs to a class of organic molecules known as **nucleic acids**. There are two

polymeric types of nucleic acids: DNA and **RNA** (**ribonucleic acid**). Working together, these molecules direct protein synthesis in cells.

The term **gene** refers to a basic unit of inheritance. The set of genes that an organism inherits is its **genotype**. An organism's characteristics (**phenotype**) are influenced by its environment but are largely determined by its genotype.

A gene can also be defined as a segment of DNA that is **transcribed** into RNA by enzymes known as **RNA polymerases**. In many cases the RNA functions as messenger RNA (**mRNA**), which is then **translated** into the amino acid sequence of a protein. Proteins have a variety of functions and are often described as doing the "work" of the cell. Many function as enzymes and catalyze metabolic reactions. Others function in maintaining cell structure, movement, membrane transport and even immunity. The protein that we will consider in this lab has a sensory function. The gene for this protein is called TAS2R38 and is one of a family of genes in humans that encode proteins involved in tasting bitter substances.

The sense of taste is mediated in part by taste receptor proteins that are found on the surface of cells on the tongue. These receptors bind to molecules (ligands) that are then perceived as having sweet, bitter or umami taste. Binding of a ligand to a taste receptor activates a signaling pathway that results in neurotransmitter release. Nerve cells associated with the taste receptor cells then transmit a signal to the brain. The brain interprets multiple taste inputs and essentially decides if something is good to eat or should be spit out. One theory to explain the ability of humans and other animals to taste bitter substances is that this helps us avoid eating toxins made by some plants. However, many harmless substances are also bitter tasting and research is ongoing regarding this question and on how the sense of taste functions in general.

The family of T2R genes in humans includes 25 to 30 different genes. Together, the proteins produced from these genes allow us to taste a variety of bitter substances. The TAS2R38 gene and the protein it encodes (designated taste receptor type 2 member 38) have been studied the most widely. This is due to the accidental discovery of their role in the ability to taste the synthetic compound phenylthiocarbamide (PTC) as described in the Introduction section of the Student Lab Instruction packet.

There are several different versions (**alleles**) of the TAS2R38 gene in humans with two alleles being most common. Inheritance of one of these, the taster allele, results in an ability to taste PTC. The other allele is the non-taster allele. The inability to taste PTC is a recessive trait and people that are homozygous for the non-taster allele do not perceive PTC as bitter tasting.

Differences in the nucleotide sequence of the TAS2R38 gene between the taster and non-taster alleles have been identified. These differences are **single nucleotide polymorphisms (SNPs)** that result in changes in the amino acid sequence of the taste receptor protein. The non-taster allele differs at three nucleotide positions compared to the taster gene. These three SNPs are inherited together and described as a **haplotype**.

DNA Extraction

Previously, we used detergent treatment to extract DNA from wheat germ cells. In this experiment, your cells will be heated to near boiling, which will break up cell and nuclear membranes, releasing DNA into solution. Using heat avoids adding chemicals that may interfere with the next step, which is amplifying the DNA using polymerase chain reaction (PCR). It also works well to inactivate enzymes called DNAses that can cut DNA into pieces. Also added to your cells are Chelex beads, which bind metal ions that can interfere with PCR.

Polymerase Chain Reaction (PCR)

The polymerase chain reaction (PCR) is a valuable tool in the molecular biology lab. Essentially, it is a procedure used to replicate a specific segment of DNA in a test tube. The following components will be included in your PCR reaction:

Template DNA (extracted from cheek cells)

Primers – short segments of DNA that will bind to a matching sequence in the TAS2R38 gene

Taq Polymerase – a heat-resistant enzyme that replicates DNA

Deoxynucleoside triphosphates (dNTPs) – monomer subunits for building new DNA strands

The PCR process has three steps that are repeated multiple times to make many copies of a specific segment of DNA. The first step is Denaturation. Here the sample is heated to 94°C, which melts the DNA, separating the strands. Next, during the Annealing step, the sample is cooled to 64°C so that the primers bind to the DNA. Finally, during the Extension step, the samples are heated to 72°C and the polymerase synthesizes new DNA strands starting where the primers are bound.

View the PCR animation on the Cold Spring Harbor DNA Learning Site:

https://dnalc.cshl.edu/resources/animations/pcr.html

3. Week One

PreLab

A. Read the Background section above and pp. S1-S2 and S6 -S8 from Student Guide Instructions. Answer the questions below:

1. What is the function of the protein coded for by the TAS2R38 gene?
2. Describe how it was discovered in the 1930s that humans differ in their ability to taste PTC.
3. Which is the dominant trait in humans for ability to taste PTC, taster or non-taster?
4. What is the likely phenotype for someone that inherits one taster allele and one non-taster allele?

Procedure: DNA Extraction

1. Read and sign the DNA collection consent form. Note that if you do not wish to collect your DNA you will be provided with anonymized class results once the lab is complete.
2. Label two 1.5 mL test tubes with the identification code given to you by your professor.
3. Follow the procedure on pp. S-6 to S-7 to isolate your DNA. Note that we will use 1.5 mL test tubes throughout the procedure. These will be placed in a lab heatblock for the 99 degree incubation.

4. Keep your sample on ice after completing the procedure.

Procedure: PCR Reactions

1. Obtain 1 PCR tube with a PCR bead. The beads have the polymerase that will copy your DNA, and nucleotides needed for this process.
2. Label the top of your PCR tube with your identification code.
3. Follow the procedure on p. S-7 to set up your PCR reaction.
4. Place your PCR tube in the carrier rack for the PCR machine while it is still on ice. Make note of the location of your tube: row and column.

Questions to be turned in next week

1. What type of cells do you expect to be found in your saline rinse (hint: think about cells we looked at in the Microscopy lab)?
2. Where is DNA located in this type of cell?
3. What procedure was used to extract DNA from cells?
4. According to the Introduction section of the Student Lab Instructions, what is the function of the Chelex beads that you mixed with your cell suspension?
5. What is the most basic function of the PCR procedure?
6. What is the role of the DNA polymerase in the PCR reaction?
7. What is the role of the primers in the PCR reaction?